

NULL 2 BUILD GUIDE

CHECKLIST

Included in the kit:

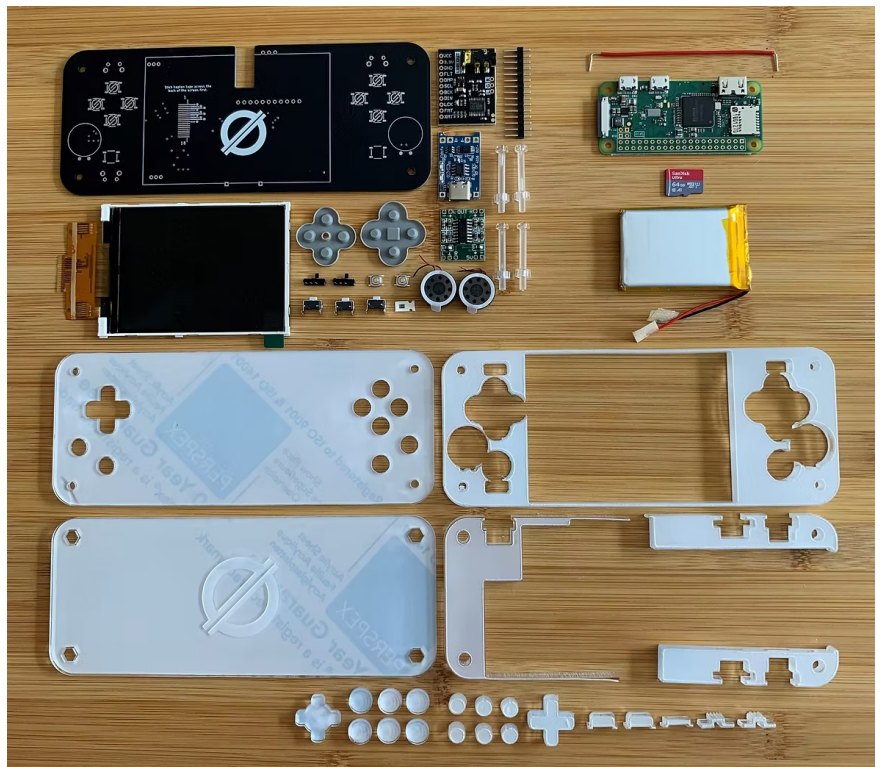
- 1 x Null 2 PCB
- 2 x SMD buttons
- 3 x Right-angled buttons
- 2 x Right-angled switches
- 1 x Set of rubber membranes
- 1 x TP4056 charger (the blue board)
- 1 x 3R9 SMD resistor
- 1 x 3.2" ili9341 screen
- 1 x PAM8403 (the green board)
- 1 x PCM5102A (the headphone board) with headers
- 2 x Speakers
- 4 x M3 20mm acrylic nuts and bolts

Not included in the kit:

- 1 x Raspberry Pi Zero W
- 1 x 1000mAh battery (5x34x50mm)
- 1 x Set of 3mm/0.118" clear acrylic parts
- 1 x Set of 5mm/0.220" clear acrylic parts
- 1 x Micro SD card

Tools/materials needed:

- Anything you need to solder
- Double-sided tape
- Kapton tape
- 1mm thick foam tape (optional)
- Masking tape
- A multimeter
- Flush cutters
- Acrylic solvent (e.g. Weld-on 4)
- A piece of single core wire
- Sandpaper



THE ACRYLIC PARTS

For the acrylic case, you will need BOTH the 3mm parts and the 5mm parts.

They can be purchased from Ponoko, Razorlab.online, or Sudomod and the design files can be found via the button above.

Razorlab.online is UK-based and offer Null 2 cases at: £35 for UK, £45 for EU, and £55 for USA, if you email them at Hello@razorlab.online

Don't use the "Make" button as it will not apply the offer, email them instead, just mention the Null 2.

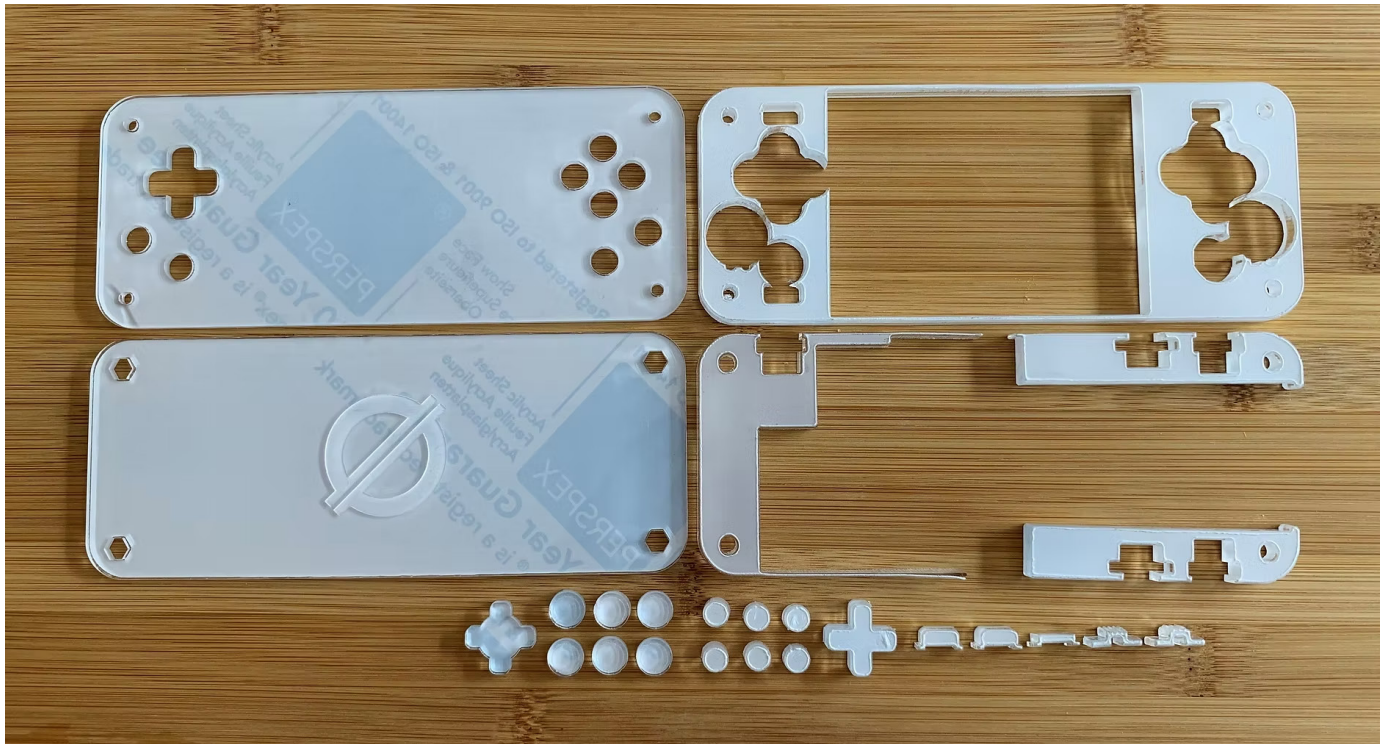
Ponoko don't offer 5mm clear acrylic, so instead use the 0.220" option. It's slightly thicker (5.6mm) but will still work.

The design files include two sets of all the buttons so if you lose, break, or ruin them, you get a second chance.

You will also get a bunch of scrap pieces, hold onto them.

Make sure you have received all of these parts and contact the laser cutters if any are missing or broken.

Some of these pieces are fragile, be careful.



These are the 3mm parts.

These are the 5mm parts.

Name your order: Null2-V1-Final



Created 24 minutes ago.

Make

Make

Your 2D designs

null2-v1-3mm-p1-final.eps

x 1

null2-v1-5mm-p1-final.eps

x 1

Copy Edit

Your 2D materials

null2-v1-3mm-p1-final.eps

Acrylic - Clear
3.0 mm
P1 - 181.0 mm long x 181.0 mm wide
Making: £9.13
Material: £1.39
Total: £10.52

Change Remove

null2-v1-5mm-p1-final.eps

Acrylic - Clear
5.0 mm
P1 - 181.0 mm long x 181.0 mm wide
Making: £16.43
Material: £1.94
Total: £18.37

Change Remove

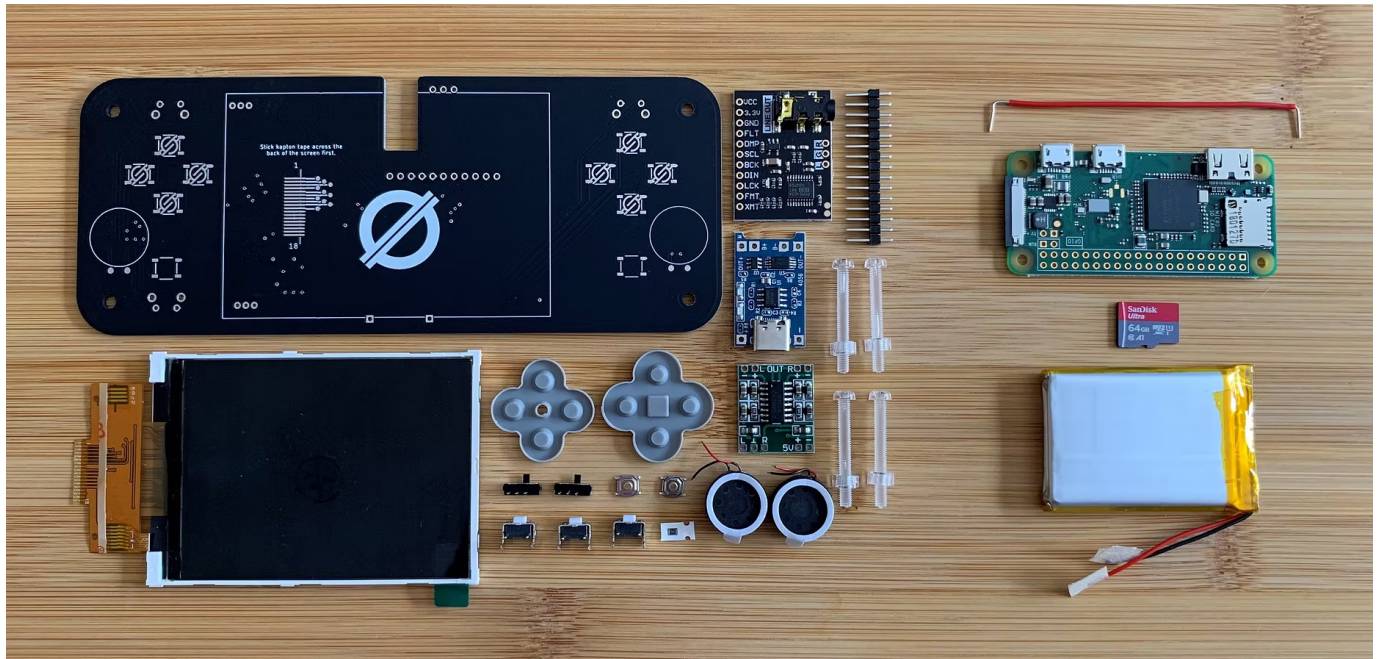
Here is an OLD screenshot of RazorLab with all the correct settings for the case. They have since changed their system in favor of a more personal one, use their live chat and mention the Null 2.

The "Null2-V1-3mm-P1-FINAL.eps" file contains all the 3mm parts for the case.
Select Acrylic - Clear, 3.0 mm, and P1.

The "Null2-V1-5mm-P1-FINAL.eps" file contains all the 5mm parts for the case.
Select Acrylic - Clear, 5.0 mm, and P1.

MAKING THE PCB

Here are all the components:



Included in the kit.

Not included in the kit.

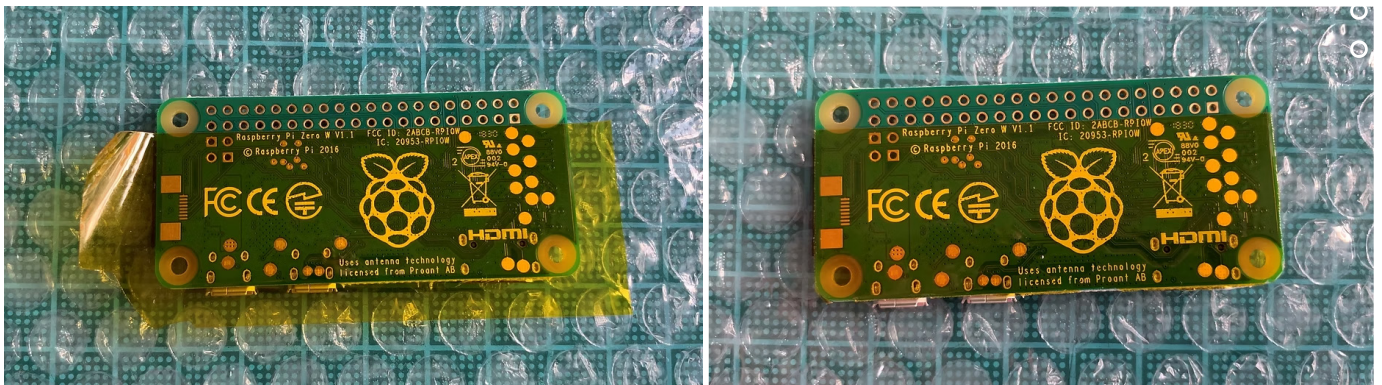
Tips:

- Take your time with everything.
- Read a little ahead in the guide or read the entire thing first.
- Do not start desoldering parts if something goes wrong. Contact me through tindie or get help on the discord.

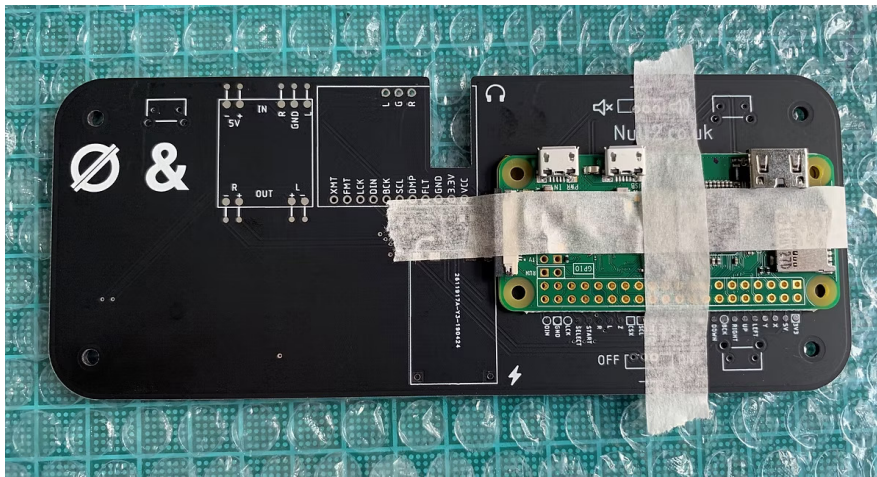
Start with the Pi. It's the hardest part to attach and the one that gives most people trouble.

If you've never surface mounted a Pi before, I recommend getting some of these and practicing first.

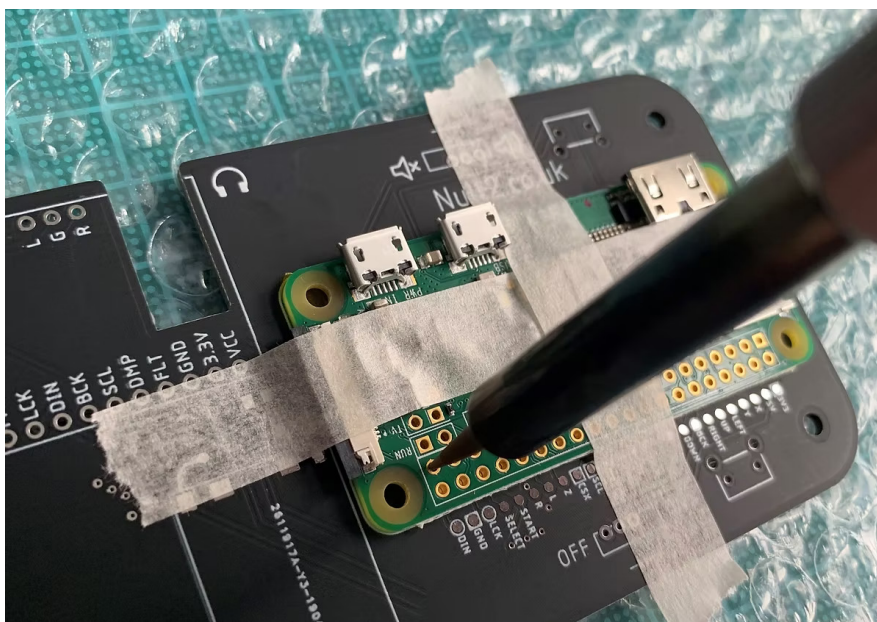
First of all, put kapton tape across the bottom of the Pi, leaving the connections along the top exposed, then trim the tape down to size:



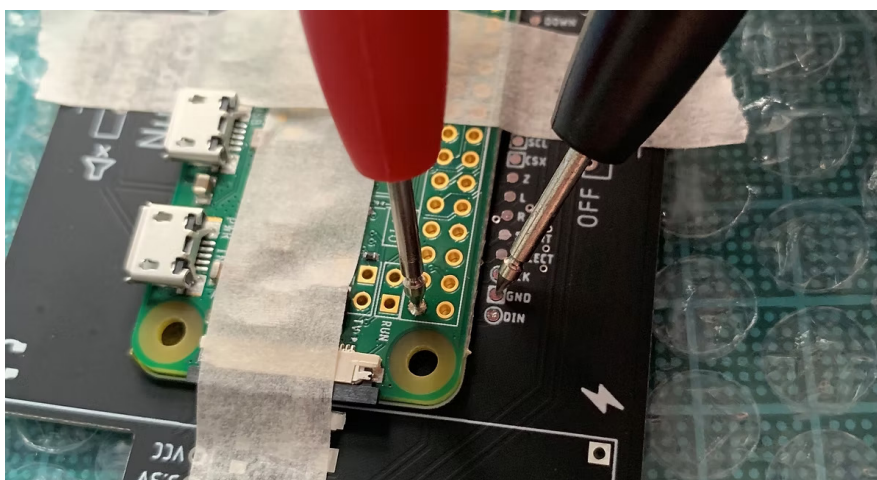
Line the Pi up with the white outline on the board and tape it securely in place:



Place the tip of the soldering iron in a hole and let it heat the pad below, introduce a **small** amount of solder, you don't want to bridge the pads below. There's a better explanation of this process and a video [here](https://www.null2.co.uk/build-guide). After you've soldered one of the pads, jump a few pads along for the next one. This will mean you are less likely to cause a bridge underneath the Pi as the solder only spreads to the heated parts.

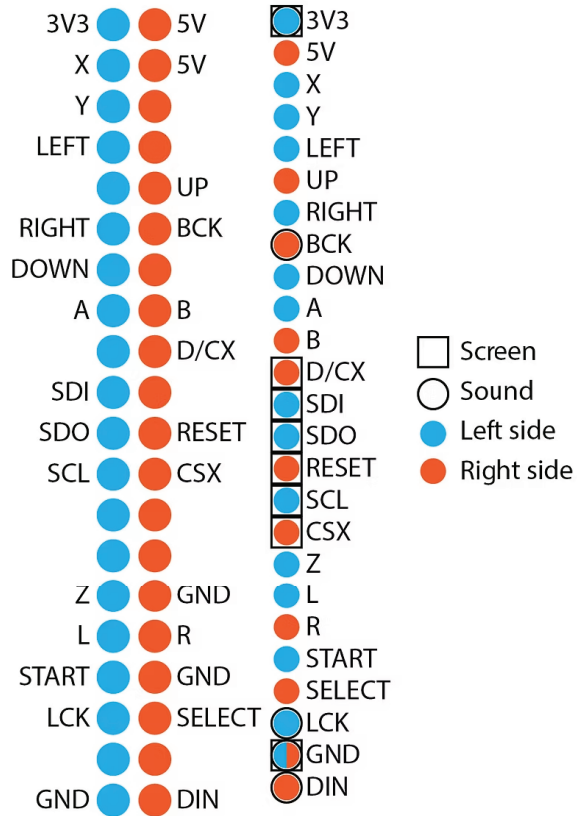
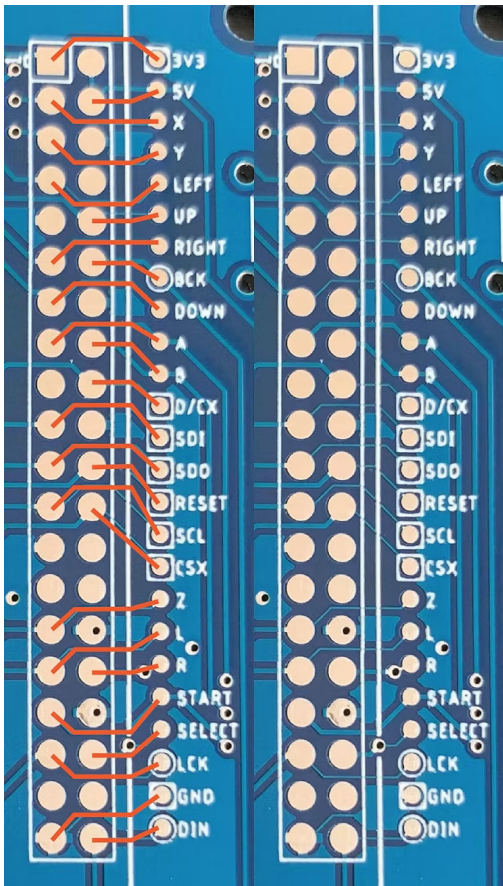


Use a multimeter to test your connections using the diagrams below. You want to make sure that there is a connection between the pad on the Pi and the test pad. You should also check that the pads on the Pi are not connected to any adjacent pads. If they are, you have likely bridged the connection under the Pi.

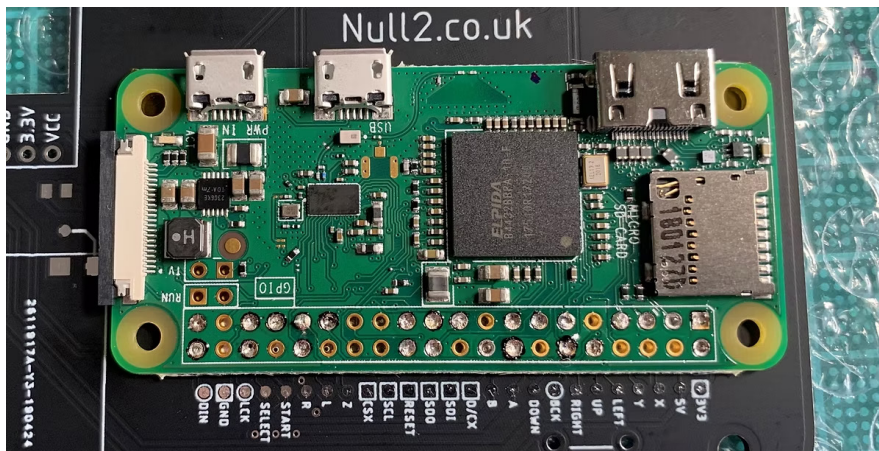


If you have any issues with your screen connections, you only need to check the pads in the **boxes**. If you have any issues with sound, you only need to check the pads in the **circles**. 3V3 and GND have both a box and a circle.

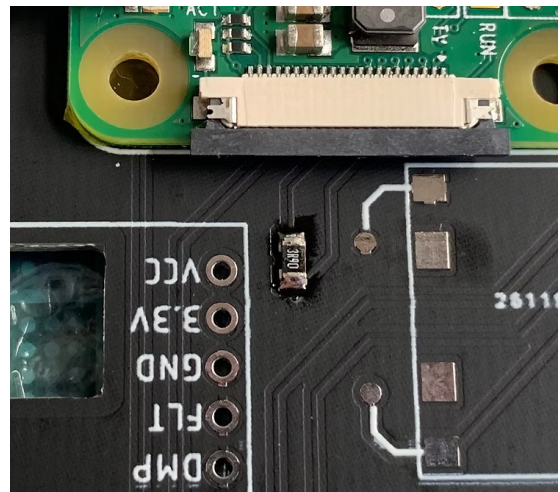
The colours are just to help you quickly identify which side of the Pi the pad you're looking for is on.



You don't need to connect all of the pads on the Pi, just the ones that have a test pad connected to them. If you choose not to solder them all, be careful when inserting and removing the SD card, as too much force may break the Pi off the board. Notice how my soldering is bowl shaped rather than bulging out, you really don't need a lot of solder to make a connection.



Next, solder on the small resistor, this can be tricky so it's best to do this now before it gets too crowded on the board. Tape one side down and solder the other end, do this for both sides:

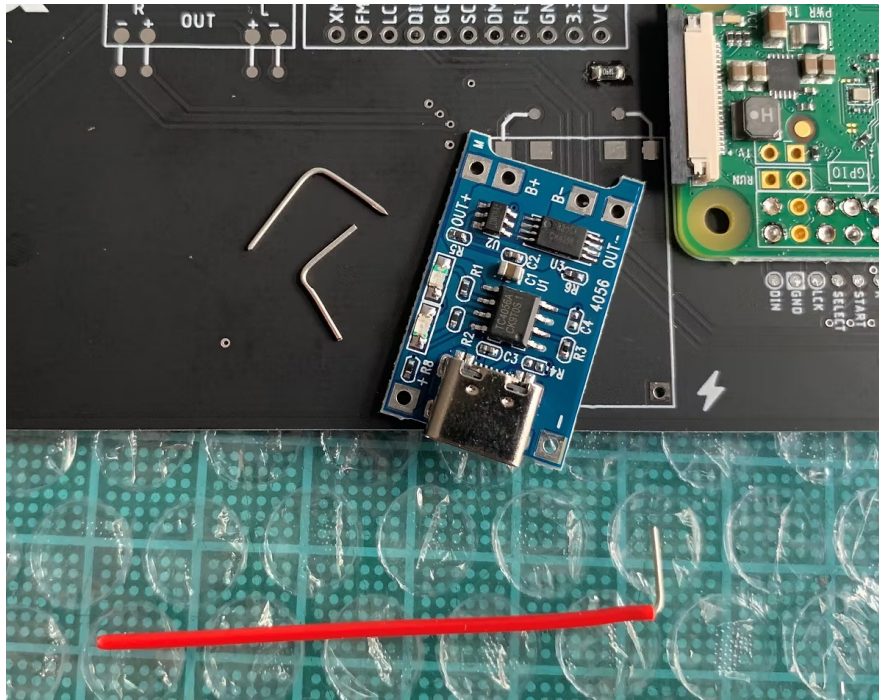


Now for the TP4056 board (also known as the blue board).

Important: Do not use a high voltage fast charger (like the one the Google Pixel uses) with the Null 2, you will damage the TP4056 module. Check your charger before using it, 5V and 2A should be safe to use.

The board included with the kit has a USB C socket, however a micro USB version is available if you prefer that.

You'll need to cut two short lengths of a piece of wire. These are used to lock the blue board onto the main PCB and stop it from shearing off when you plug and unplug the charger.

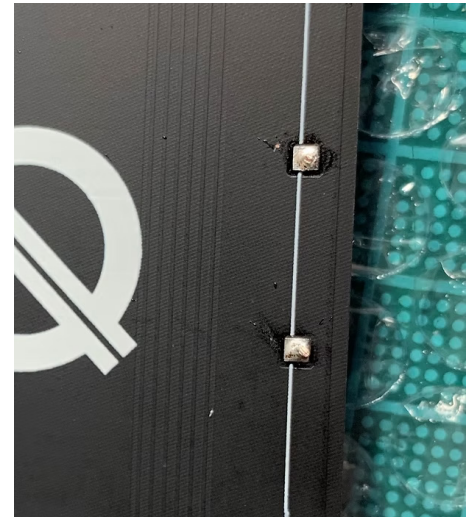
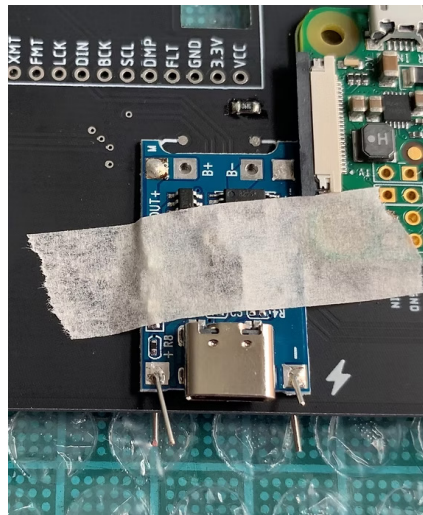


Tape the board down and place the wire pieces through the holes beside the socket.

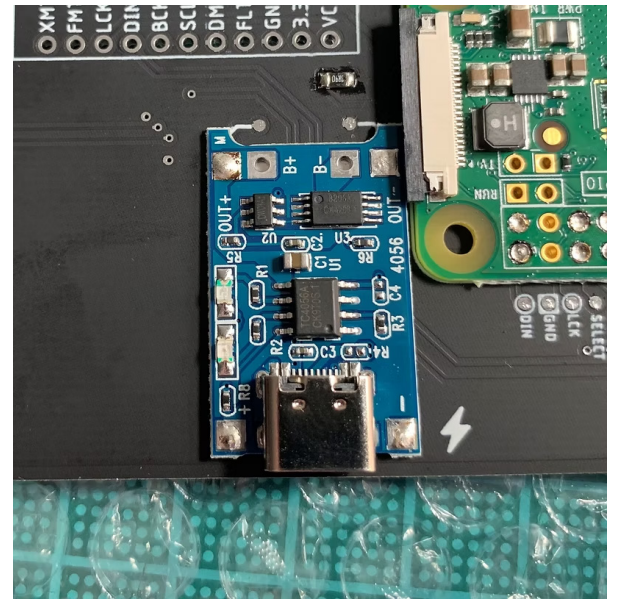
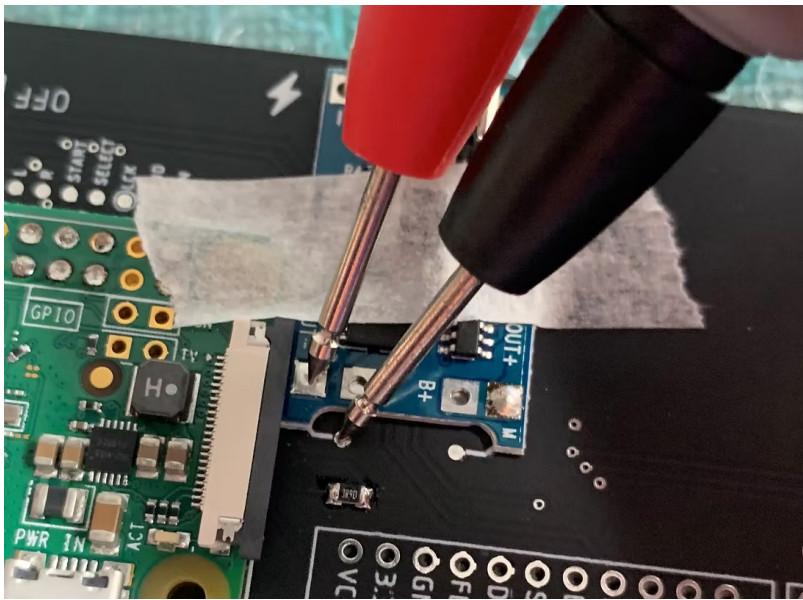
Solder the wires on both sides of the board and cut off the ends.

The wires will heat up when you solder them so make sure you do this on a safe surface.

Solder the **top right and top left** pads (marked OUT+ and OUT-) in the same way that you soldered the Pi.

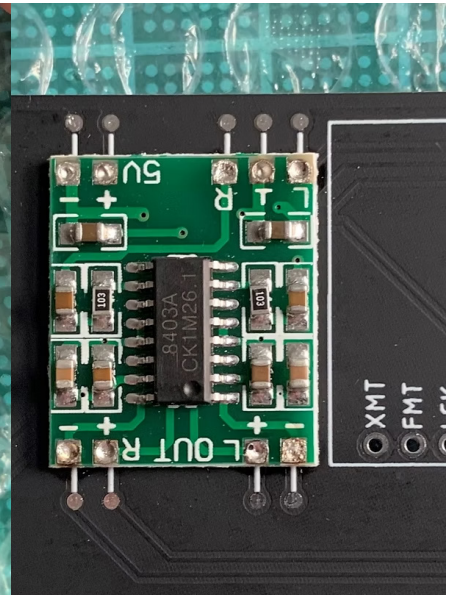
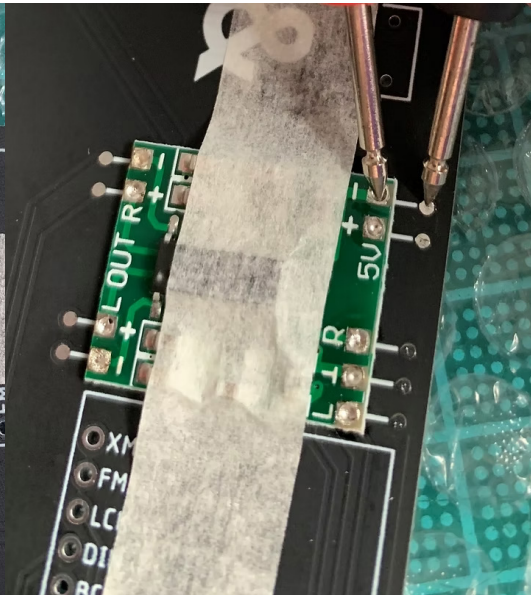
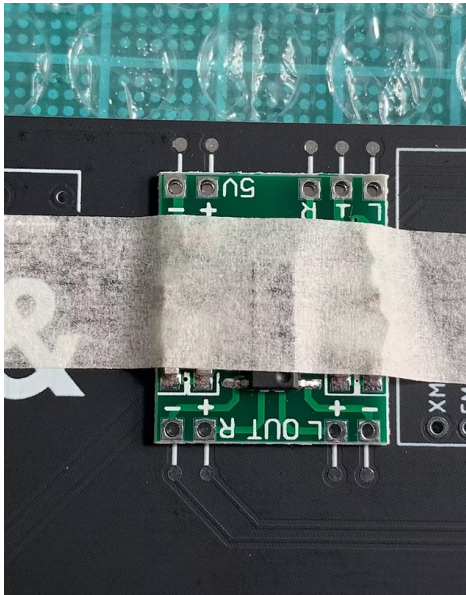


Test the top corner connections for continuity with the pads just above them. These connections are important so press the blue board down and try wiggling it a little to ensure that the connection doesn't break. Don't try to break it, just make sure it's secure.



Now the PAM8403 board (also known as the green board).

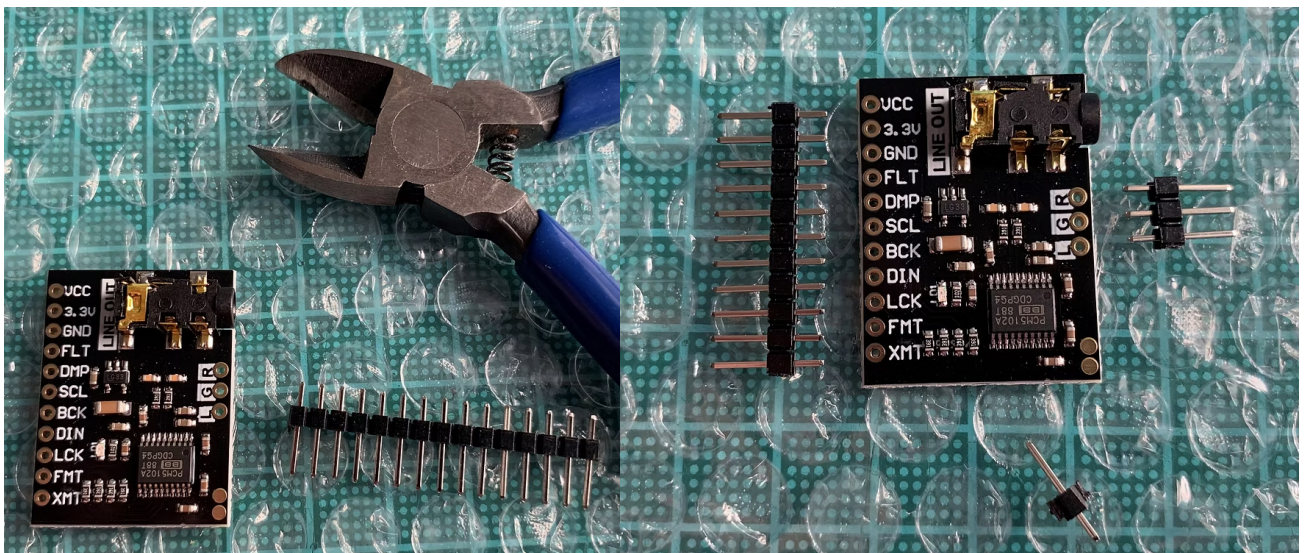
Tape it in place. Solder it the same way as the Pi. Test your connections between the pads and the test pads above and below (follow the white lines).

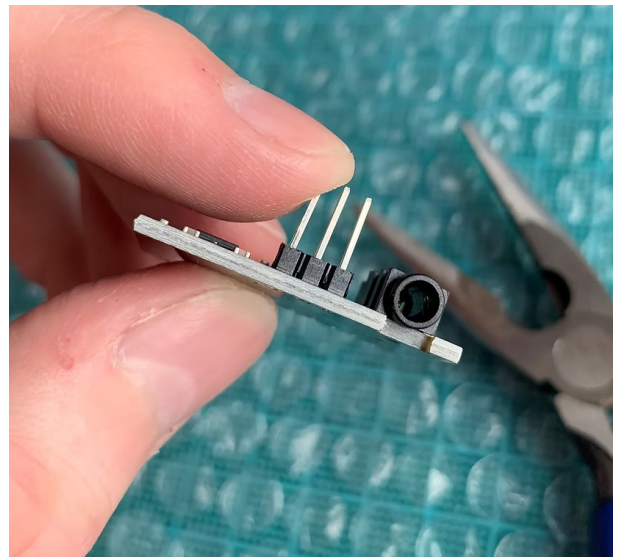
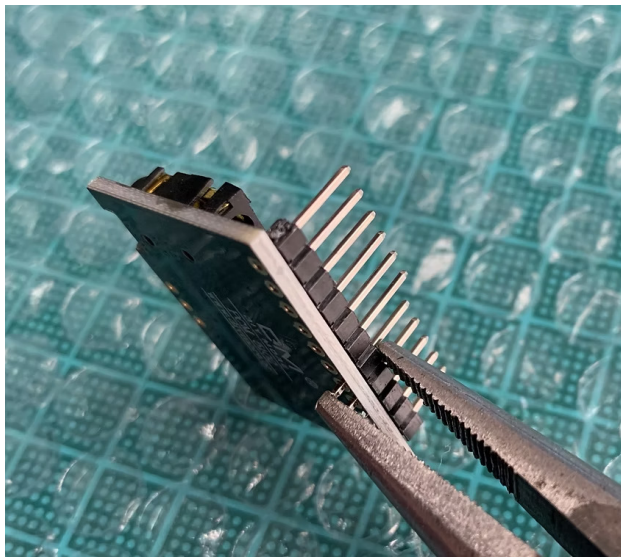


Now the PCM5102A board (also known as the headphone board). **This one is tricky.**

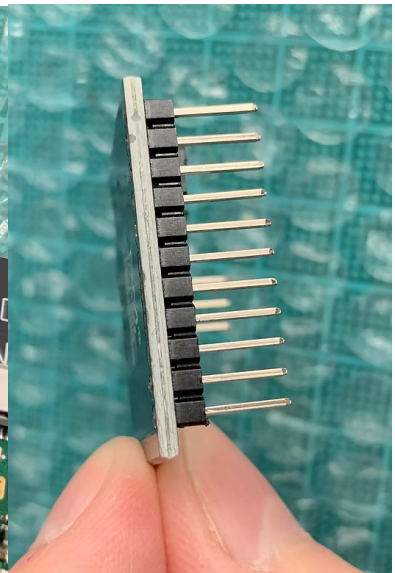
Cut the header into a row of 3 and a row of 11.

Place the header into the headphone board and use pliers to move the pins one by one so that they sit flush against the flat side of the board.

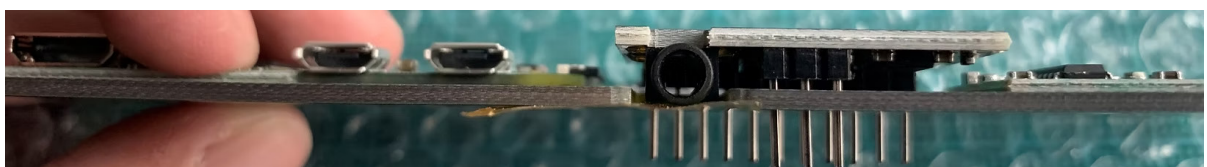
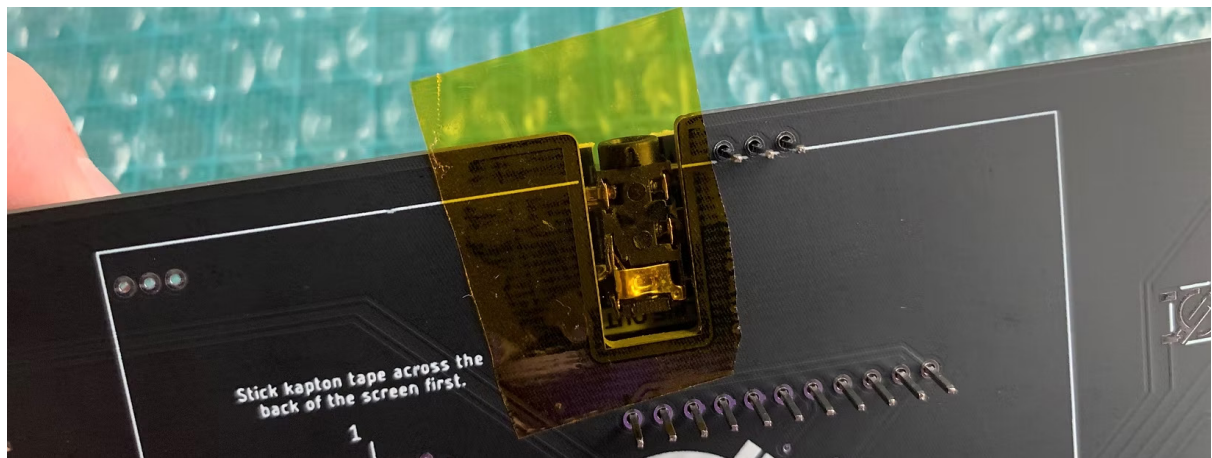




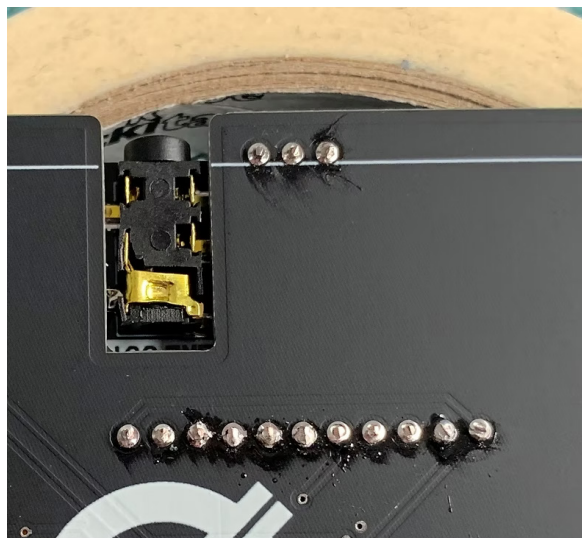
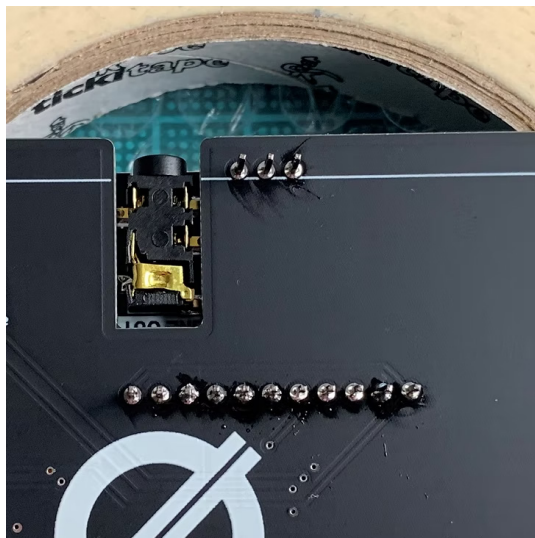
Use the Null 2 PCB as a jig to hold them in the right place. Solder each of the pins to JUST the headphone board. Make sure you don't use too much solder. If you used too much and the solder bulges out, remove the excess with solder wick or a solder sucker. If these connections stick out, they will hit the inside of the case. The pins will get hot when soldered so make sure you do this on a safe surface.



The headphone board needs to be connected to the Null 2 PCB at a certain height. Too high and it will hit the back of the case, too low and you won't be able to get your headphones in. To make sure it's in the right place, put some kapton tape across gap in the Null 2 PCB to stop the headphone socket from going through fully. The distance between the rear face of the PCB and the top face of the headphone board should be ≤ 5 mm.

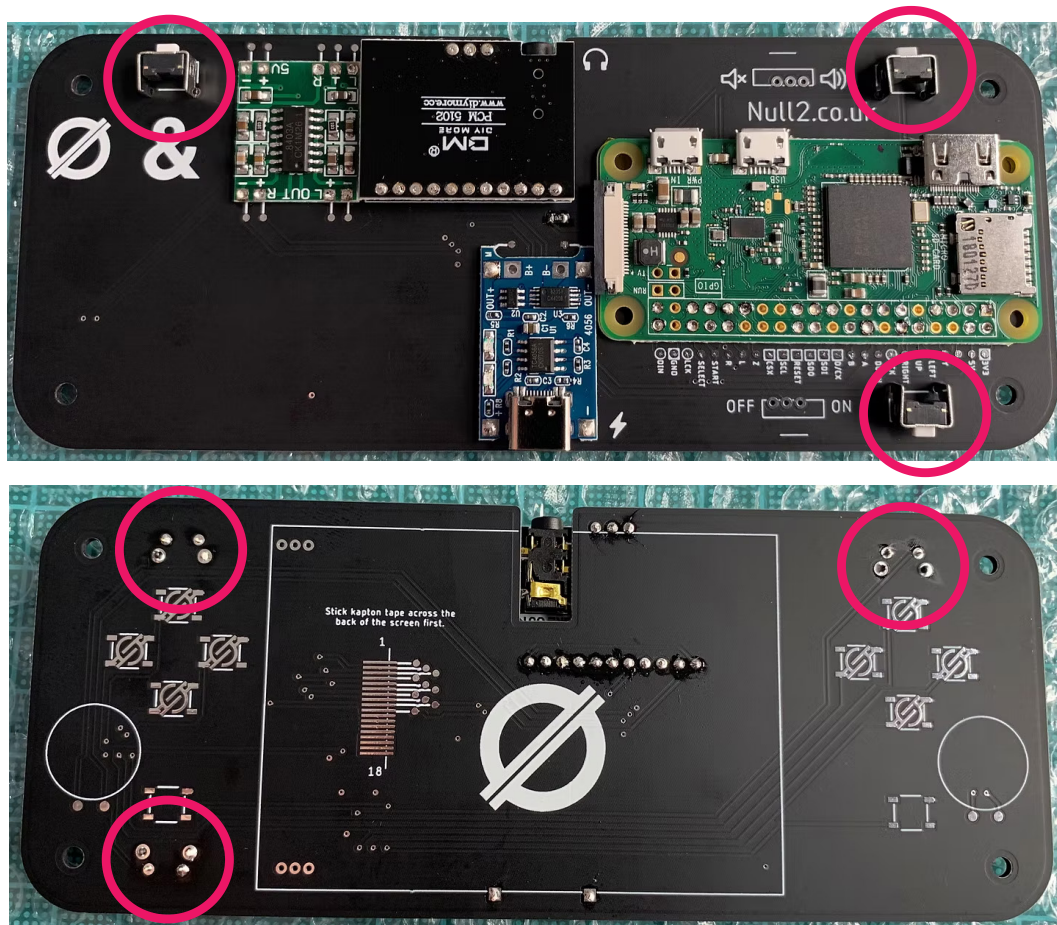


Solder the pins to the Null 2 PCB with the board held at the correct height and cut them down as short as you can using flush cutters, you don't want them to press into the back of the screen. Be careful, the pins like to fly off/at your face so put your thumb over them when you snip.



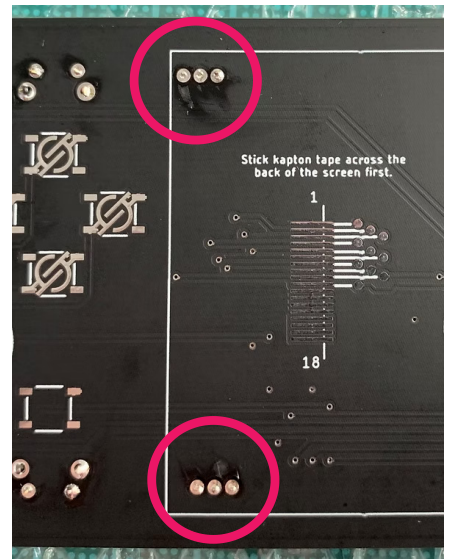
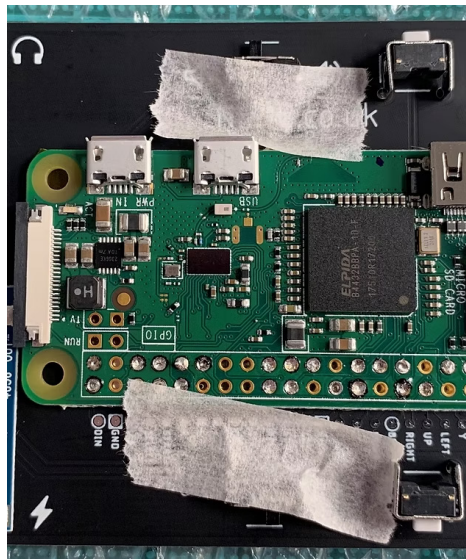
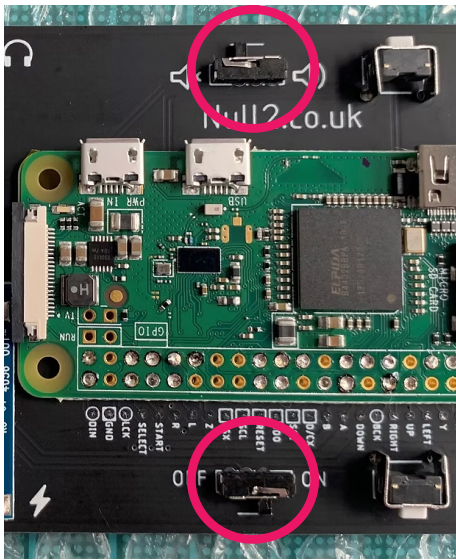
Now the right-angled buttons for the L, R, and Z buttons (Z is the 'System' button on the bottom edge of the board).

Place the buttons into their respective places on the BACK of the PCB, tape them down if needed. Turn the board over and solder them to the Null 2 PCB. You don't need to cut the legs off.



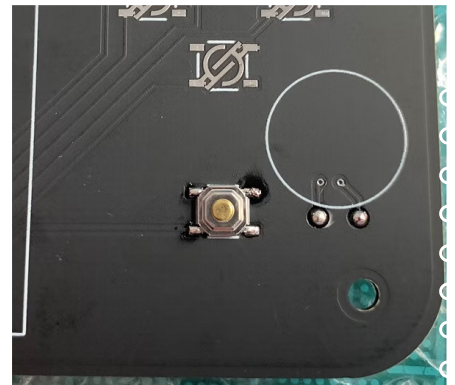
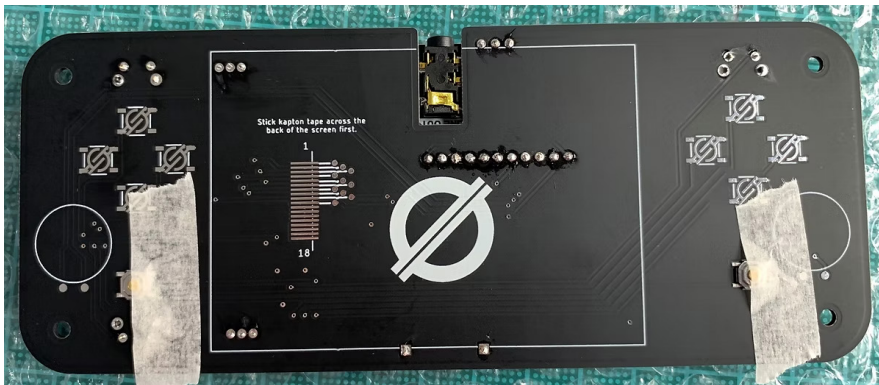
Now the mute and power switches.

Place the switches into their respective places on the BACK of the PCB, tape them down (déjà vu). Turn the board over and solder them to the Null 2 PCB.



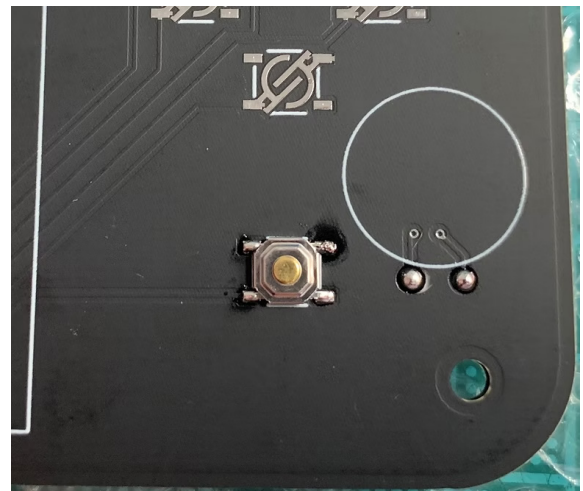
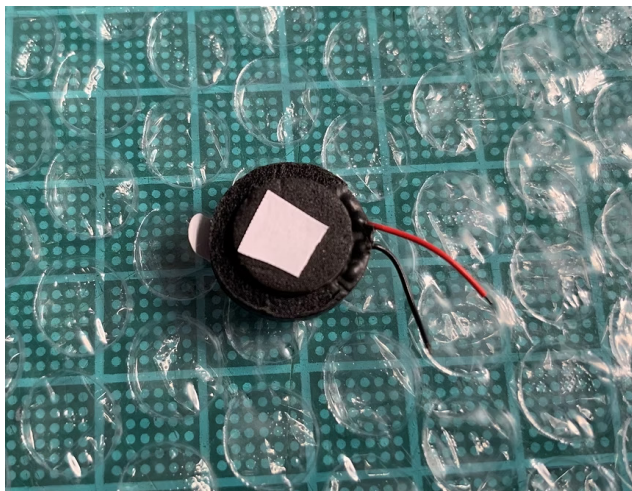
Now the Start and Select buttons.
Tape them in place. Solder them on.

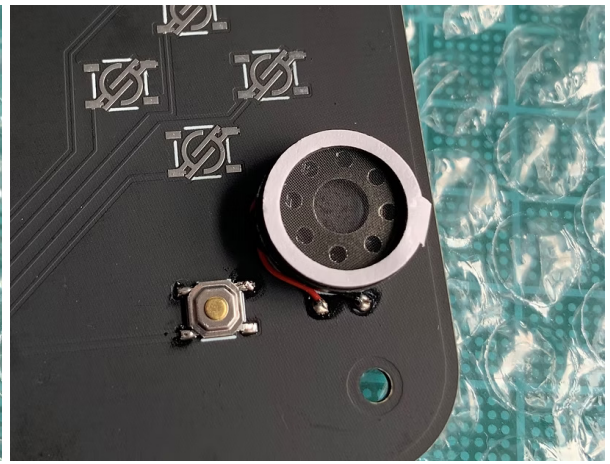
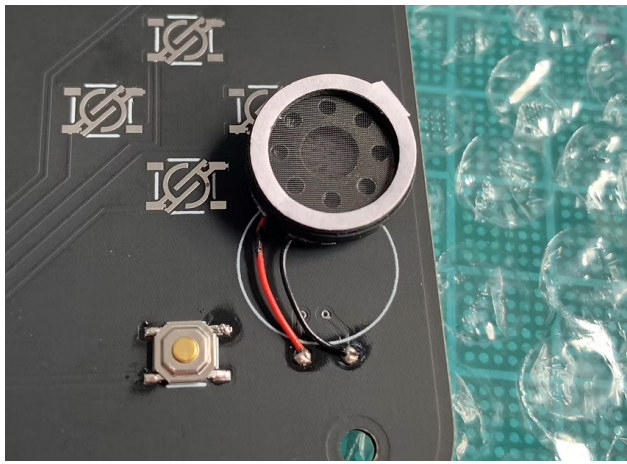
If you want to use these tactile buttons instead of the rubber membranes, you can buy some here and solder them on in the same way. This is an optional thing and not necessary.



Now the speakers.

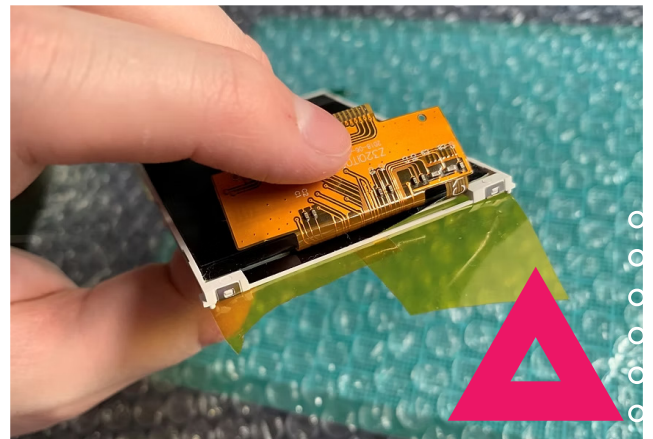
Put some double-sided tape on the back of the speakers. Heat the pads and put a blob of solder onto each one. Place a leg of the speaker into a blob of solder and hold it there while the solder cools down. This can be a bit annoying but it will work. The red wire goes to the positive pad on the left and the black wire goes to the negative pad on the right. Stick them down, wrapping the wires around the outside and tucking them underneath.





Now the screen.

Put kapton tape across the back of the screen to prevent shorts. Being very careful of the ribbon, cut the tape down to size. **DO NOT CUT THE RIBBON.**



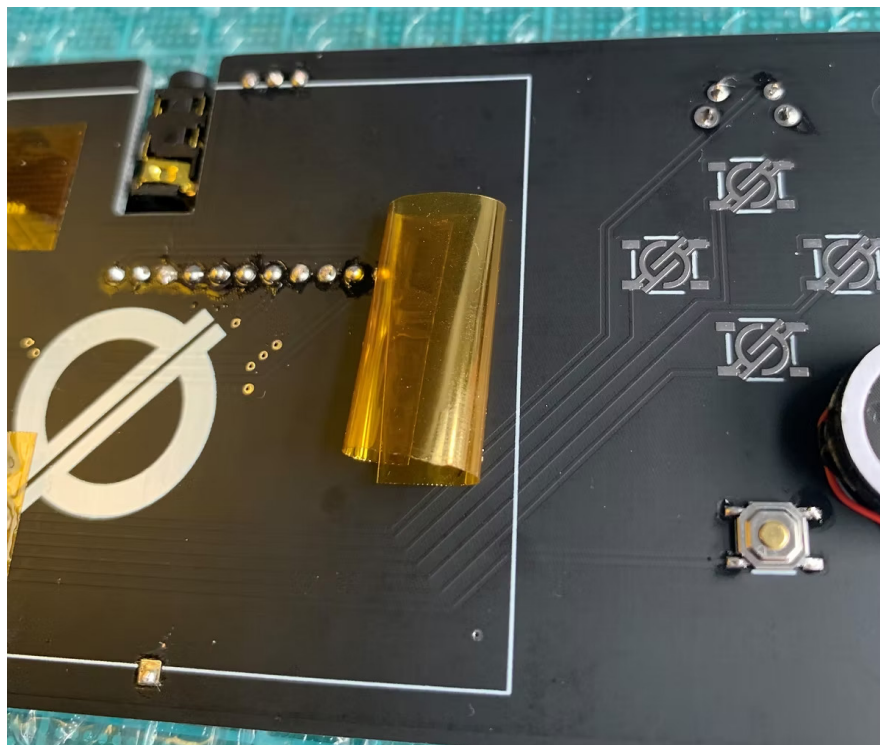
Align the edge of the ribbon with the white line on the board and make sure the pads line up. Tape it in place above and below. Solder the ribbon on, this video might be helpful. The bottom 4 pads aren't connected to anything so you can consider them a sort of practice run if you've never done this before.

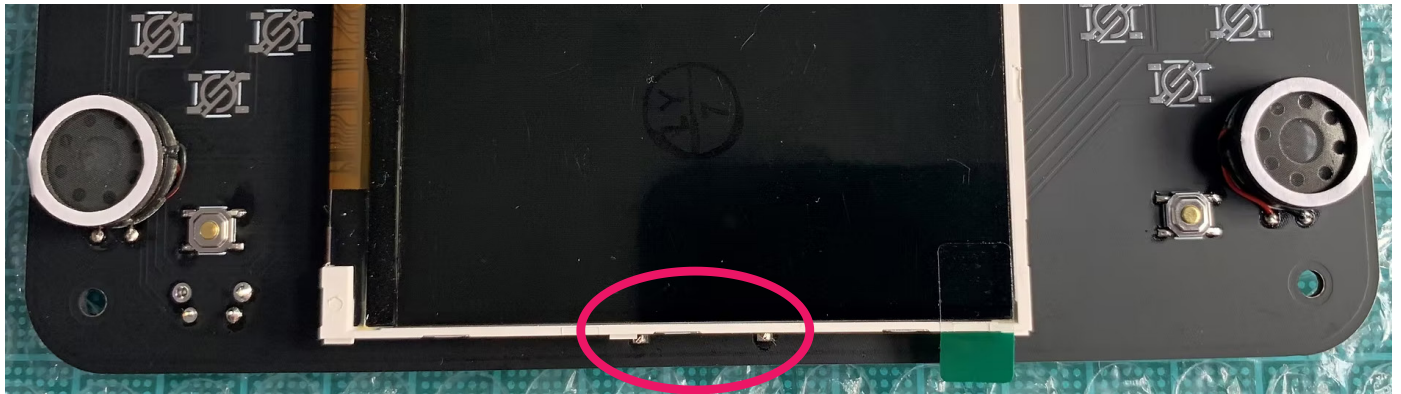
If you test your connections with a multimeter, test pads 9 and 10 (the lower two test pads) will say they're connected if the resistor on the back is correctly soldered, this is not a problem.

If you think something is wrong, do not remove the screen, join the Discord server and ask for help.



Make a loop with some kapton tape and press the screen down, keeping it aligned with the white outline. The metal frame of the screen must not touch the 2 metal contacts at the bottom or the 3 metal contacts at the top so it's important to get it lined up properly. If either of these connections touch the metal frame, put kapton tape along the corresponding edge of the screen.

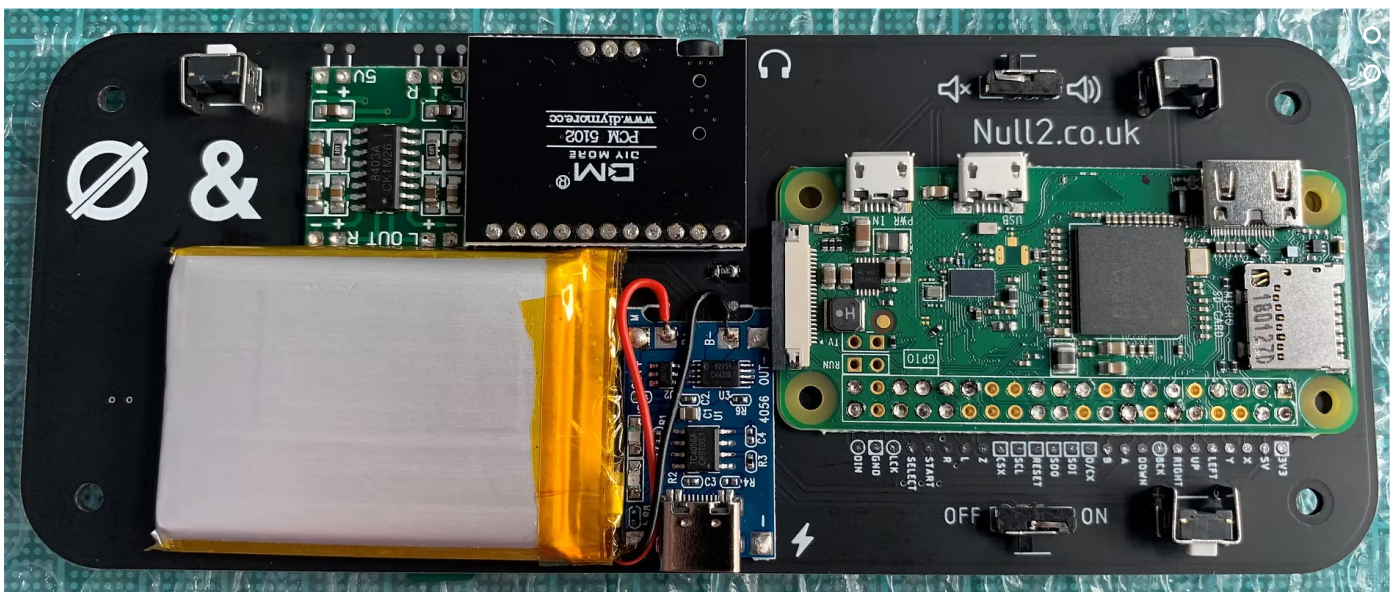
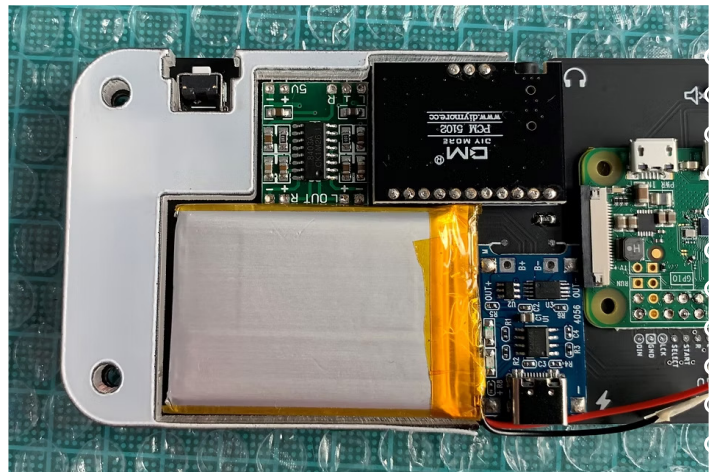
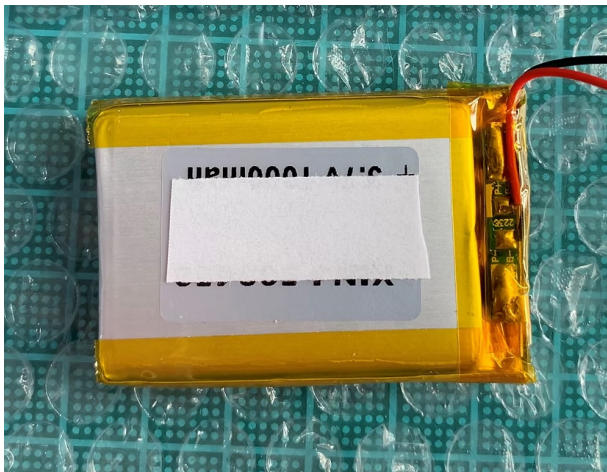




Finally the battery.

Put a piece of double-sided tape across the back of the battery. Solder the red wire (positive) directly to the pad on the blue board marked B+ and solder the black wire (negative) to the pad marked B-.

Use the case to check alignment and stick the battery down.



Soldering finished!

You can put the SD card in and check everything behaves the way it's supposed to. The first time the image boots it will take slightly longer as it expands the file system for your SD card. Shut it down by holding the System button and pressing B, then wait for the activity light on the Pi to stop flashing before flicking the power switch to avoid corrupting your card.